

Session 3: Joins & Basic Queries - Solutions

Use Case 1: Highest Confirmed Cases using INNER JOIN

SQL Query

```
SELECT c.country_name, cc.confirmed_cases
FROM covid_cases cc
INNER JOIN countries c
ON cc.country_id = c.country_id
WHERE cc.report_date = '2021-01-01'
ORDER BY cc.confirmed_cases DESC
LIMIT 1;
```

Explanation

Joins covid_cases with countries, filters a specific date, sorts by confirmed cases descending and returns the top country.

Use Case 2: LEFT JOIN COVID Deaths and Vaccination Data

SQL Query

```
SELECT d.country_id, c.country_name, d.total_deaths, v.vaccination_status
FROM covid_deaths d
INNER JOIN countries c ON d.country_id=c.country_id
LEFT JOIN covid_vaccines v
ON d.country_id=v.country_id;
```

Explanation

Displays all countries with death counts. Vaccination status is shown where available; otherwise NULL.

Use Case 3: Total Deaths by Continent

SQL Query

```
SELECT ct.continent_name,
SUM(cc.total_deaths) AS total_deaths
FROM covid_cases cc
INNER JOIN countries c
ON cc.country_id=c.country_id
INNER JOIN continents ct
ON c.continent_id=ct.continent_id
GROUP BY ct.continent_name
ORDER BY total_deaths DESC;
```

Explanation

Aggregates total deaths for each continent using SUM() and GROUP BY.

Use Case 4: Average New Deaths Per Day

SQL Query

```
SELECT report_date,
AVG(new_deaths) AS avg_daily_deaths
FROM covid_deaths
GROUP BY report_date
ORDER BY report_date;
```

Explanation

Calculates average new deaths for each date and sorts chronologically.

Use Case 5: Highest Infection Rates**SQL Query**

```
SELECT c.country_name,  
confirmed_cases,  
population,  
(confirmed_cases*100.0/population) AS infection_rate  
FROM covid_cases cc  
INNER JOIN countries c  
ON cc.country_id=c.country_id  
ORDER BY infection_rate DESC;
```

Explanation

Computes infection rate percentage and ranks countries from highest to lowest.